

MAVRA learning and maintenance

1-2

learning

3-7

maintenance

other outcome

relearn

1 Learn

built-in agent (LLM)

question: April return amount
SUM(sr_return_amt) ... d_moy = 4
answer 3,133,115.63, verified ✓
validated before publishing ✓



2 Publish v1

the definition holds

SUM(sr_return_amt)
rules: one row per return,
date key unique,
dates complete; learning SQL

3 Normal write

late returns appended

store_returns 12 → 13
rechecked: still holds ✓
date_dim same: cached ✓
stays v1

✗ No unique repair

e.g. a full backup copy

sr_source='primary' ✓
sr_source='backup' ✓
both work: cannot tell
retired until relearned

4 Breaking write

a status row per return

store_returns 13 → 14
one row per return: fails ✗
v1 retired
unfixed: answer doubles

5 Find a repair

try filters on few-value columns

sr_status='done' ✓
sr_status='applied' loses rows ✗
exactly one works ✓

6 Regression test

rerun on the learning-time data

learning SQL 3,133,115.63
with the filter 3,133,115.63
same: fix accepted ✓

7 Publish v2

new version, same metric

v2 = v1 + filter
sr_status='done'
SQL naming v1: rejected
waiting requests get v2

a later write

two work, or none